

# Math Bootcamp

Day 2 — More calculus and existence

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Math Camp 2026

# How to use these slides

**Main deck:** definitions, recipes, examples, and practice.

**Appendix and proofs** (marked on relevant slides): **for interest only** — optional recap for those already comfortable with the math.

**Not required for Math Camp.** Focus on the main slides and exercises unless you want the extra detail.

# Don't make your partner disappointed

 新浪网

 司天钦Gosta  
26-02-03 11:50 微博认证: 美食博主

女生男生都超级扣分的行为，快来看看自己中了几条

1. 不定积分结果不加常数C
2. 定积分换元时忘记更换积分上下限
3. 滥用洛必达法则
4. 求复合函数导数时遗漏链式法则项
5. 判别级数敛散性误用判别法
6. 计算行列式或逆矩，目属代数余子式的符号
7. 混淆偏导数与全导数
8. 分部积分法中 $du$ 和 $dv$ 选择不当
9. 在加减运算中，误用等价无穷小
10. 解非齐次微分方程时遗漏特解

## 2.1 Explicit vs. implicit relationships

**Explicit:** one variable is written directly as a function of others.

$$y = g(x) \quad \text{e.g.} \quad p^* = \frac{\alpha - c}{b + d}.$$

**Implicit:** variables are linked by an equation  $F(x, y) = 0$ .

$$F(x, y) = 0 \quad \text{e.g.} \quad E(p, \alpha) = 0 \quad (\text{market clearing}).$$

Often we **know**  $F$  but cannot solve for  $y = g(x)$  in closed form. The IFT says: under regularity conditions,  $y$  is still locally a function of  $x$ , and we can compute  $\frac{dy}{dx}$  without solving explicitly.

## 2.1 What problem does the IFT solve?

**Comparative statics question.** How does an endogenous variable respond when an exogenous parameter shifts?

- Equilibrium price  $p^*(\alpha)$  when demand depends on shifter  $\alpha$
- Optimal choice  $x^*(w)$  from a first-order condition  $F(x, w) = 0$
- Equilibrium objects defined by  $F(x, \theta) = 0$  without closed-form solutions

**Pattern:** an equilibrium condition  $F(x, \theta) = 0$  defines  $y^*(\theta)$  implicitly. We want  $\frac{dy^*}{d\theta}$  — the IFT delivers this from partials of  $F$ .

## 2.1 One-variable IFT — statement and recipe

Let  $F(x, y)$  be  $C^1$  near  $(x_0, y_0)$  with  $F(x_0, y_0) = 0$ . If  $F_y(x_0, y_0) \neq 0$ , then locally there is a unique  $g$  such that

$$y = g(x), \quad F(x, g(x)) = 0, \quad g(x_0) = y_0,$$

and

$$\boxed{g'(x) = \frac{dy}{dx} = -\frac{F_x(x, y)}{F_y(x, y)}} \quad \text{on the curve } F(x, y) = 0.$$

**Recipe.** Totally differentiate  $F(x, y) = 0$ :

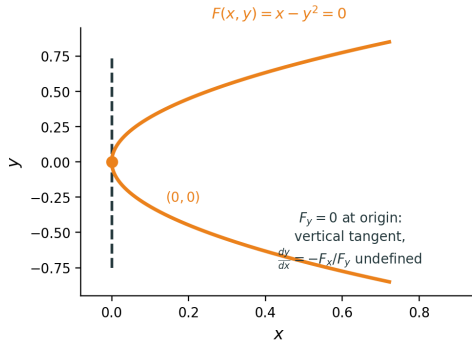
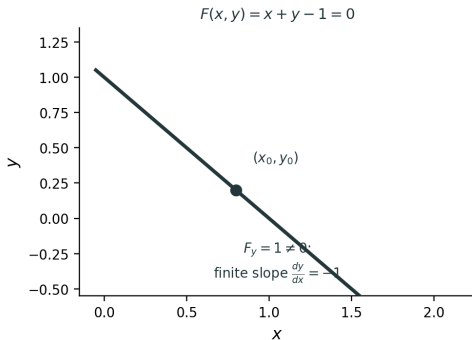
$$0 = dF = F_x dx + F_y dy \quad \Rightarrow \quad \frac{dy}{dx} = -\frac{F_x}{F_y} \quad \text{when } F_y \neq 0.$$

Several variables (**for interest only**): [appendix](#).

## 2.1 Why $F_y \neq 0$ ? — regularity condition

**Left:**  $F_y \neq 0$  at  $(x_0, y_0)$  — tangent has finite slope; locally  $y = g(x)$ . **Right:**  $F_y = 0$  at  $(0, 0)$  — **vertical tangent**;  $\frac{dy}{dx} = -F_x/F_y$  is undefined.

Why the IFT needs  $F_y \neq 0$



## 2.1 — Comparative statics practice (no closed form)

Together

**Nonlinear demand:**  $D(p, \alpha) = \alpha e^{-p}$ ,  $S(p) = p$ ,  $\alpha > 0$ .

**No elementary closed form for  $p^*(\alpha)$ .**



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5.  $\alpha = 2$ :  $2e^{-p} = p$  gives  $p^* \approx 0.85$ , so  $\frac{dp^*}{d\alpha} \approx 0.23$ .

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No closed form for  $p^*(\alpha)$  — IFT still works.

## 2.1 Applications — motivation

**You just used the IFT recipe.** Write  $E(p, \theta) = 0$ , compute  $E_p$  and  $E_\theta$ , then  $\frac{dp^*}{d\theta} = -\frac{E_\theta}{E_p}$  — even when  $p^*(\theta)$  has no closed form.

**Why these examples?** They are the standard micro patterns behind that recipe:

- **Demand shifter  $\alpha$ :** when does equilibrium price move up or down?
- **Linear market:** IFT matches the answer from solving  $E = 0$  explicitly
- **Tax  $t$ :** same formula, different parameter — who bears the tax?

**Goal.** Connect the abstract formula to comparative statics you will see in economics.

## 2.1 Application — setup

**Market clearing.** Quantity demanded equals quantity supplied:

$$D(p, \alpha) = S(p),$$

where  $p$  is price and  $\alpha > 0$  is a **demand shifter** (e.g. consumer income, taste index, population).

Define excess demand

$$E(p, \alpha) = D(p, \alpha) - S(p).$$

An equilibrium price  $p^*(\alpha)$  satisfies

$$E(p^*(\alpha), \alpha) = 0.$$

**Question:** when  $\alpha$  rises, does equilibrium price  $p^*$  rise or fall? Use  $\frac{dp^*}{d\alpha} = -\frac{E_\alpha}{E_p}$ .

## 2.1 Application — linear example

Linear demand and supply:

$$D(p, \alpha) = \alpha - bp, \quad S(p) = c + dp, \quad b, c, d > 0.$$

Excess demand  $E(p, \alpha) = D(p, \alpha) - S(p) = \alpha - c - (b + d)p$ . Equilibrium  $p^*(\alpha)$  satisfies  $E(p^*(\alpha), \alpha) = 0$ .

Partials:  $E_p = -(b + d) \neq 0$ ,  $E_\alpha = 1$ .

IFT:

$$\frac{dp^*}{d\alpha} = -\frac{E_\alpha}{E_p} = \frac{1}{b + d} > 0.$$

## 2.1 Application — check by explicit solution

From  $E(p^*(\alpha), \alpha) = 0$ :

$$p^*(\alpha) = \frac{\alpha - c}{b + d}.$$

Differentiate explicitly:

$$\frac{dp^*}{d\alpha} = \frac{1}{b + d},$$

matching the IFT formula  $-\frac{E_\alpha}{E_p}$ .

**When IFT is essential:** nonlinear  $D(p, \alpha)$  or  $S(p)$  —  $p^*(\alpha)$  may have no closed form, but

$$\frac{dp^*}{d\alpha} = -\frac{E_\alpha}{E_p} \quad \text{at } p = p^*(\alpha)$$

still gives the local comparative static (as in the practice problem).

## 2.1 Application — economic interpretation

**Chain rule.** Since  $E(p^*(\alpha), \alpha) = 0$  for all  $\alpha$ ,

$$0 = \frac{d}{d\alpha} E(p^*(\alpha), \alpha) = \underbrace{E_\alpha(p^*, \alpha)}_{\text{direct}} + \underbrace{E_p(p^*, \alpha) \frac{dp^*}{d\alpha}}_{\text{indirect via } p^*}.$$

So  $\frac{dp^*}{d\alpha} = -\frac{E_\alpha}{E_p}$  when  $E_p \neq 0$ .

**Two effects of higher  $\alpha$  on excess demand (in words).**

1. **Direct effect:** holding  $p$  fixed, higher  $\alpha$  raises demand  $\Rightarrow E$  goes **up** ( $E_\alpha > 0$ ).
2. **Indirect effect:** equilibrium price adjusts; higher  $p$  makes  $E$  **lower** ( $E_p < 0$ ).

**Why must  $p^*$  rise?** The direct effect alone would create excess demand at the old price. Price rises until the indirect effect offsets it and restores  $E = 0$ . With  $E_\alpha > 0$  and  $E_p < 0$ , this requires  $\frac{dp^*}{d\alpha} > 0$ .

## 2.1 Example — tax incidence (economics)

**Setup.** Demand  $D(p) = \alpha - bp$ , supply  $S(p) = c + dp$  ( $b, d > 0$ ). A per-unit tax  $t$  on **sellers**: buyers pay  $p$ , sellers receive  $p - t$ .

Clearing:  $D(p) = S(p - t)$ , so  $E(p, t) = \alpha - bp - c - d(p - t) = 0$  and  $p^*(t)$  is implicit.

Partials:  $E_p = -(b + d)$ ,  $E_t = d$ . **IFT:**

$$\frac{dp^*}{dt} = -\frac{E_t}{E_p} = \frac{d}{b + d} > 0.$$

**Incidence shares (per \$1 of tax).** Buyer price rises by  $\frac{dp^*}{dt} = \frac{d}{b + d}$ ; seller net price falls by  $\frac{b}{b + d}$ .

**Interpretation (elasticity).** In this linear market, larger  $b$  means more elastic demand and larger  $d$  means more elastic supply. The tax is borne **more heavily by the less elastic side**: if demand is relatively inelastic (small  $b$ ), buyers pay most of the tax; if supply is relatively inelastic (small  $d$ ), sellers pay most.



## 2.2 EVT on a closed interval $[a, b]$

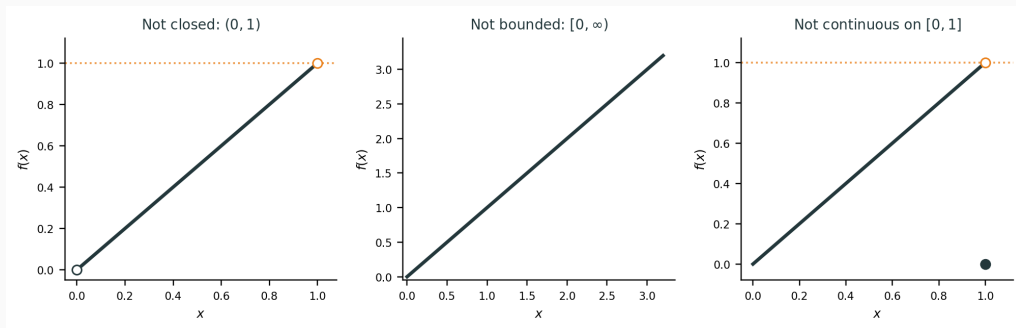
**Extreme Value Theorem (EVT).** If  $f$  is **continuous** on the **closed** interval  $[a, b]$ , then  $f$  attains a **maximum** and a **minimum** on  $[a, b]$ .

So  $\exists c, d \in [a, b]$  such that

$$f(c) \leq f(x) \leq f(d) \quad \text{for all } x \in [a, b].$$

$[a, b]$  is **closed and bounded** — a compact set in  $\mathbb{R}$ .

## 2.2 When EVT fails



Need continuity on a **closed, bounded** domain (or compact set more generally).

## 2.2 Open and closed sets

**Open set** (intuition): every point has a small neighborhood still inside the set.

**Closed set** (intuition): contains its **boundary**; limits of sequences in the set stay in the set.

**Economic examples.**

- Budget  $\{x : p \cdot x \leq B\}$  — **closed** (includes the budget line).
- Strict budget  $p \cdot x < B$  — **not closed** (boundary excluded).
- Reservation price set  $P = \{p : p < \bar{p}\}$  — not closed;  $\sup P = \bar{p}$  may not be attained.

On  $\mathbb{R}^1$ :  $[a, b]$  is closed and bounded;  $(a, b)$  is open (not closed). The failure figure above: not closed, not bounded, or not continuous.

## 2.2 Compact sets and Weierstrass EVT

In  $\mathbb{R}^n$  (camp level): a set is **compact** when it is **closed and bounded** (Heine–Borel, stated).

**Weierstrass Extreme Value Theorem.** If  $f$  is continuous on a compact set  $K \subset \mathbb{R}^n$ , then  $f$  attains a **maximum** and **minimum** on  $K$ .

**Why economists care.**

compact feasible set  $K$  + continuous objective  $f \Rightarrow$  optimum **exists**

(preview of Day 4 optimization).

Proof: **appendix (for interest only)**.

## 2.2 Example — existence of a utility maximizer

**Setup (consumer problem).** Prices  $p_1, p_2 > 0$ , income  $w > 0$ , budget set

$$\mathcal{B} = \{(x_1, x_2) \in \mathbb{R}_+^2 : p_1 x_1 + p_2 x_2 \leq w\}.$$

1.  $\mathcal{B}$  is **closed** (weak inequality  $\leq$ ) and **bounded** (e.g.  $0 \leq x_i \leq w/p_i$ ), hence **compact** in  $\mathbb{R}^2$ .
2. Suppose utility  $u(x_1, x_2)$  is **continuous** on  $\mathcal{B}$ .
3. By Weierstrass EVT,  $u$  attains a **maximum** on  $\mathcal{B}$ .

**Conclusion.** An optimal consumption bundle  $x^* \in \mathcal{B}$  **exists** (existence theorem for the consumer problem; first-order conditions come on Day 4).

**Not uniqueness.** Weierstrass gives at least one maximizer, not a unique one (e.g.  $u(x_1, x_2) = x_1 + x_2$  has many optima on  $\mathcal{B}$ ). Uniqueness needs extra conditions (e.g. strictly concave / strictly quasiconcave  $u$ ; Day 4).

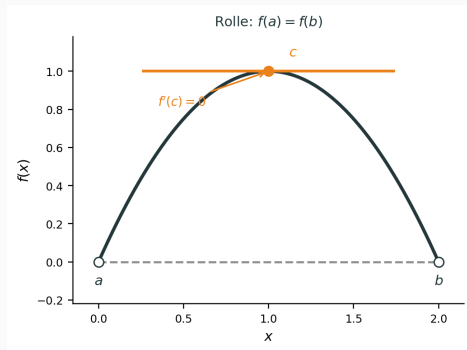
## 2.2 Rolle's theorem

**Rolle's theorem.** If  $f$  is continuous on  $[a, b]$ , differentiable on  $(a, b)$ , and  $f(a) = f(b)$ , then  $\exists c \in (a, b)$  with  $f'(c) = 0$ .

for interest only

**Proof.** If  $f$  is constant on  $[a, b]$ , any  $c \in (a, b)$  works. Otherwise, by EVT,  $f$  has a maximum or minimum on  $[a, b]$ . Since  $f(a) = f(b)$ , this extremum occurs at some **interior** point  $c \in (a, b)$ . At an interior peak or trough, the tangent is horizontal, so  $f'(c) = 0$ .  $\square$

Special case of the MVT below with  $f(a) = f(b)$ .



## 2.2 Mean Value Theorem

**Mean Value Theorem (MVT).** If  $f$  is continuous on  $[a, b]$  and differentiable on  $(a, b)$ , then  $\exists c \in (a, b)$  such that

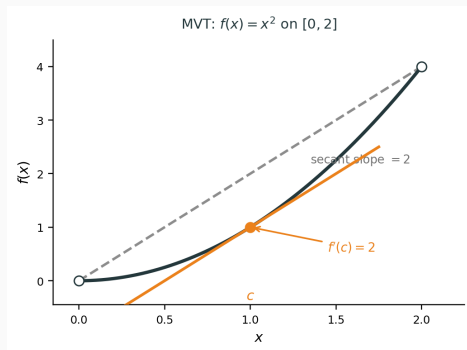
$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

**Picture:** some interior tangent is **parallel** to the secant through  $(a, f(a))$  and  $(b, f(b))$ .

**Example.**  $f(x) = x^2$  on  $[0, 2]$ : secant slope = 2, so  $f'(c) = 2c = 2$  gives  $c = 1$ .

for interest only

**Proof idea:** define  $g(x) = f(x) - m(x - a)$  with  $m = \frac{f(b) - f(a)}{b - a}$ ; then  $g(a) = g(b)$ , Rolle gives  $f'(c) = m$ .



## 2.2 Example — labor and output (economics)

**Setup.** Output  $f(L)$  from labor  $L$ ;  $f$  continuous on  $[L_1, L_2]$ , differentiable on  $(L_1, L_2)$ .

**MVT.**  $\exists L^* \in (L_1, L_2)$  with

$$f'(L^*) = \underbrace{\frac{f(L_2) - f(L_1)}{L_2 - L_1}}_{\text{average product gain over } [L_1, L_2]} .$$

At some interior employment level, **marginal product** = **average product** of the change.



## 2.2 MC and AC — what the crossover means

**Cost example.**  $C(q) = q^2 + 4$  (fixed cost 4, rising marginal cost).

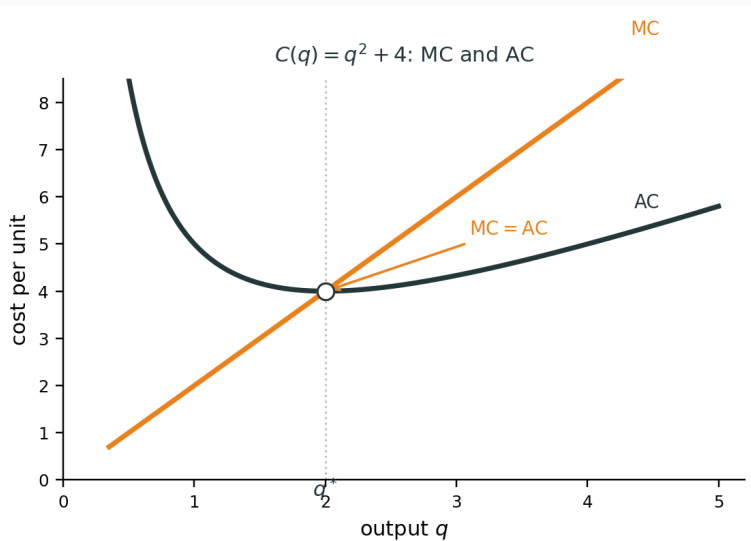
$$MC(q) = C'(q) = 2q, \quad AC(q) = \frac{C(q)}{q} = q + \frac{4}{q}.$$

**MVT link.** On  $[0, \bar{q}]$ ,  $\exists q^* \in (0, \bar{q})$  with  $MC(q^*) = \frac{C(\bar{q}) - C(0)}{\bar{q}} = \bar{q}$ : marginal cost somewhere equals the **average incremental cost** of the first  $\bar{q}$  units. (If  $C(0) = 0$ , this is  $AC(\bar{q})$ .)

**Why MC crosses AC at the minimum of AC.** Minimizing  $AC(q) = C(q)/q$  gives  $MC(q) = C(q)/q$  (FOC)  $\Rightarrow$  curves cross at  $q^* = 2$ .

- For  $q < q^*$ :  $MC < AC \Rightarrow$  an extra unit pulls the average **down**.
- For  $q > q^*$ :  $MC > AC \Rightarrow$  an extra unit pulls the average **up**.
- At  $q^*$ : the last unit costs exactly the average — **efficient scale** (lowest cost per unit).

## 2.2 MC and AC — graph



## 2.3 Integral as area and accumulation

**Riemann idea.** For  $f \geq 0$  on  $[a, b]$ ,

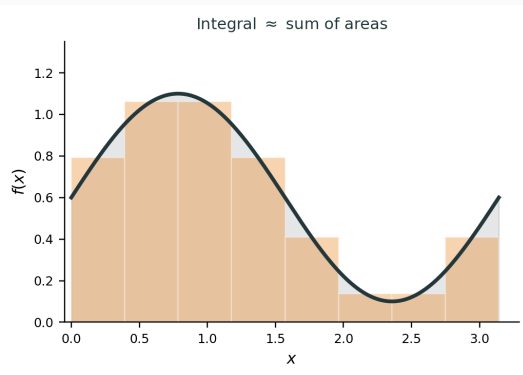
$$\int_a^b f(x) dx \approx \sum_{i=1}^n f(x_i^*) \Delta x$$

is the **signed area** under the curve / total accumulation.

**Definite vs indefinite.**

- Definite:  $\int_a^b f(x) dx$  is a number
- Indefinite:  $\int f(x) dx = F(x) + C$  is a family of antiderivatives

Econ: total cost, consumer surplus, probability mass.



## 2.3 Riemann vs. Lebesgue — for interest

Not required for Math Camp.

Two ways to define  $\int f$ .

### Riemann (this camp)

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Chop the  $x$ -axis into small pieces

Sum (height)  $\times$  (width)

Good for hand calculations

Area under curve picture

### Lebesgue (grad / probability)

---

Chop the  $y$ -axis into levels

Sum (level)  $\times$  (measure of set where  $f$  hits that level)

Integrates more pathological functions

Foundation for modern probability

**Contrast in one sentence.** Riemann slices **vertically**; Lebesgue slices **horizontally** by value.

## 2.3 Lebesgue integral — for interest

Not required for Math Camp.

**When Riemann fails.** Riemann needs the function to be “well behaved” on intervals: too many jumps can prevent upper and lower sums from converging to the same number.

**Classic example (not Riemann integrable).** Dirichlet function on  $[0, 1]$ :

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Every interval contains rationals and irrationals, so upper and lower Riemann sums never agree — no Riemann integral.

**Lebesgue handles this.** It integrates by **level sets**, so a “sparse” jump set like  $\mathbb{Q}$  does not block integration the way Riemann’s interval partitions do.

## 2.3 Fundamental Theorem of Calculus

**Part I.** If  $F'(x) = f(x)$ , then

$$\int_a^b f(x) dx = F(b) - F(a).$$

**Part II.** If  $f$  is continuous and  $F(x) = \int_a^x f(t) dt$ , then  $F'(x) = f(x)$ .

Differentiation and integration are inverse operations (MVT underlies many proofs).

## 2.3 Integration by parts

**From the product rule.** For differentiable  $u(x)$  and  $v(x)$ ,

$$(uv)' = u'v + uv' \quad \Rightarrow \quad uv' = (uv)' - u'v.$$

Integrate both sides:

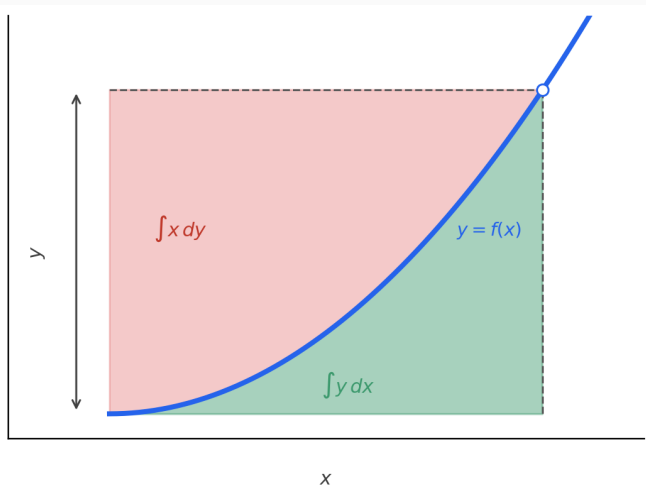
$$\int u v' dx = uv - \int u' v dx \quad \text{or} \quad \boxed{\int u dv = uv - \int v du}.$$

On  $[a, b]$  with  $u'v$  integrable,

$$\int_a^b u(x)v'(x) dx = \left[ u(x)v(x) \right]_a^b - \int_a^b u'(x)v(x) dx.$$

Pick  $u$  and  $dv$  so the new integral  $\int v du$  is simpler.

## 2.3 Integration by parts (graph)





## 2.3 Integration by parts — example

**Example.**  $\int x e^x dx$ .

Let  $u = x$ ,  $dv = e^x dx$ , so  $du = dx$ ,  $v = e^x$ :

$$\int x e^x dx = x e^x - \int e^x dx = x e^x - e^x + C = e^x(x - 1) + C.$$

**Check.**  $\frac{d}{dx} [e^x(x - 1)] = e^x(x - 1) + e^x = x e^x$ . ✓

## 2.3 PMF and CMF — discrete example (dice)

**Setup.** Fair die:  $X \in \{1, 2, 3, 4, 5, 6\}$ .

**Probability mass function (PMF):**  $p(k) = P(X = k) = \frac{1}{6}$  for  $k = 1, \dots, 6$ .

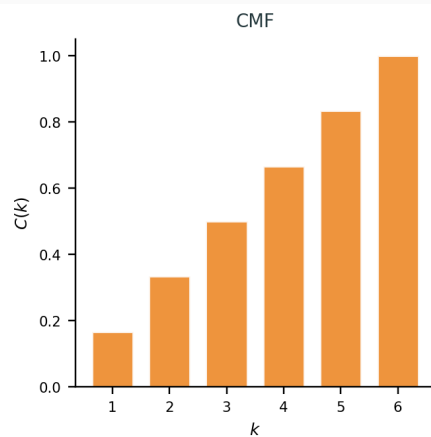
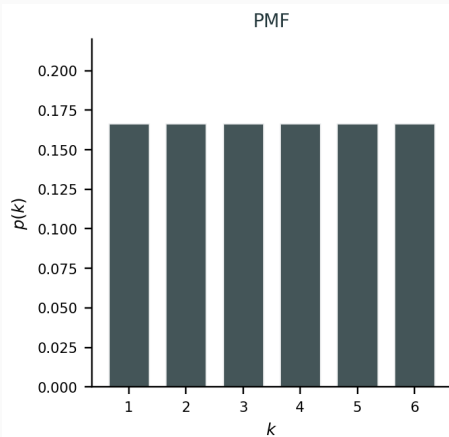
**Event (sum rule):**  $P(2 \leq X \leq 4) = p(2) + p(3) + p(4) = \frac{1}{2}$ .

**Cumulative mass function (CMF):**  $C(k) = P(X \leq k) = \sum_{j \leq k} p(j)$ , defined only at the **outcomes**  $k = 1, \dots, 6$  (not a continuous curve in  $x$ ).

Example:  $C(3) = p(1) + p(2) + p(3) = \frac{1}{2}$ .

Discrete: **PMF/CMF** and **sums**. Continuous (next slides): **PDF/CDF** and **integrals**.

## 2.3 PMF and CMF — dice (graph)



## 2.3 PDF and CDF — continuous case

Let  $X$  be a continuous random variable.

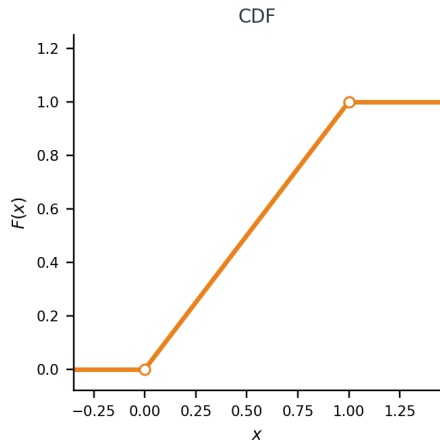
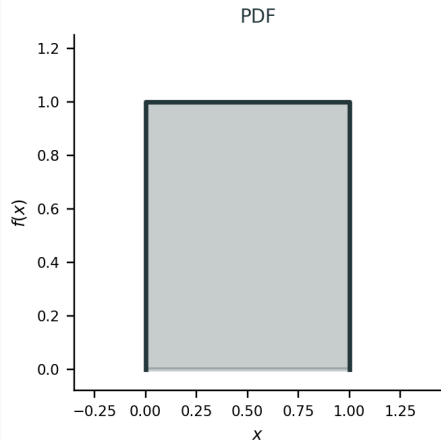
**Probability density function (PDF):**  $f(x) \geq 0$  and  $\int_{-\infty}^{\infty} f(x) dx = 1$ . Then

$$P(a \leq X \leq b) = \int_a^b f(x) dx.$$

**Cumulative distribution function (CDF):**  $F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$ ; a **continuous** curve in  $x$  (not jumps on a finite outcome list).

**FTC:** if  $f$  is continuous, then  $F'(x) = f(x)$ .

## 2.3 PDF and CDF — Unif(0, 1) (graph)



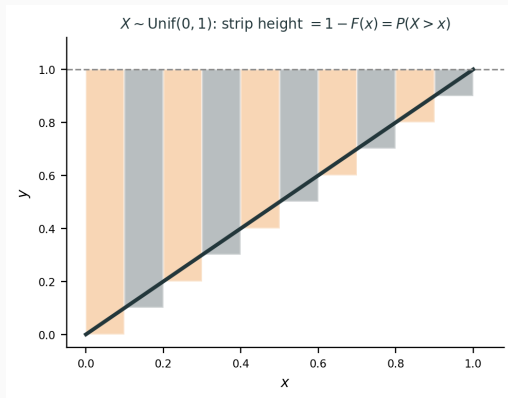
## 2.3 Expectation from the CDF

For a **non-negative** continuous  $X$  with PDF  $f$  and CDF  $F$ ,

$$\mathbb{E}[X] = \int_0^{\infty} xf(x) dx = \int_0^{\infty} (1 - F(x)) dx.$$

**Intuition.** For  $X \sim \text{Unif}(0, 1)$ , each vertical strip has height  $1 - F(x)$ ;  $\mathbb{E}[X]$  is their total area.

Derivation via integration by parts (**for interest only**):  
[appendix](#).



## 2.4 Leibniz rule — fixed limits

**Setup.**  $F(\theta) = \int_a^b f(x, \theta) dx$  with fixed  $a, b$ .

**Leibniz rule (fixed limits).** When regularity holds,

$$\frac{dF}{d\theta} = \int_a^b \frac{\partial f}{\partial \theta}(x, \theta) dx.$$

**Regularity (enough for this camp).**

- $\frac{\partial f}{\partial \theta}(x, \theta)$  exists
- $f$  and  $\frac{\partial f}{\partial \theta}$  are **continuous** in  $(x, \theta)$

Then you may swap  $\frac{d}{d\theta}$  and  $\int_a^b$ .

**Idea.** Only the integrand moves with  $\theta$ ; the limits  $a, b$  stay fixed.

When regularity fails (**for interest only**): [appendix](#).

## 2.4 Leibniz rule — variable limits

**Setup.**  $F(\theta) = \int_{a(\theta)}^{b(\theta)} f(x, \theta) dx.$

**Leibniz rule (variable limits).** When  $a(\theta)$ ,  $b(\theta)$  are differentiable and regularity holds,

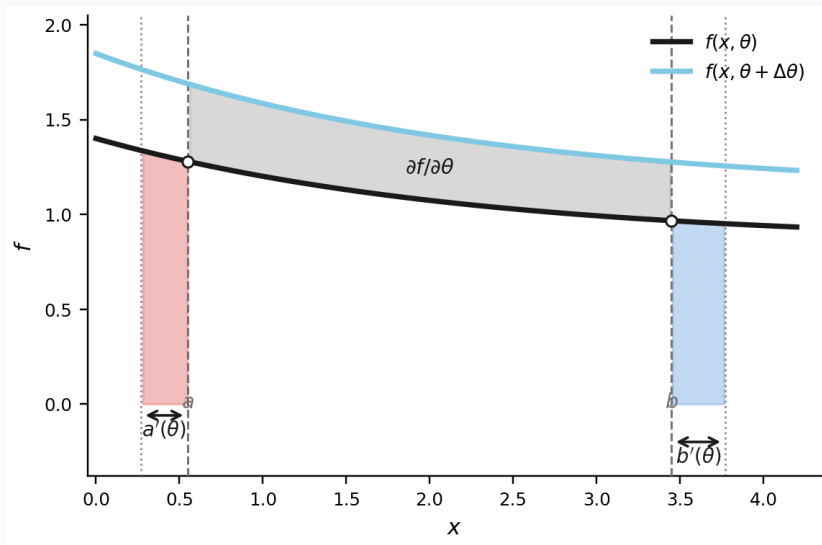
$$\frac{dF}{d\theta} = f(b(\theta), \theta) b'(\theta) - f(a(\theta), \theta) a'(\theta) + \int_{a(\theta)}^{b(\theta)} \frac{\partial f}{\partial \theta}(x, \theta) dx.$$

**Three parts.** (i) area added/lost at the **upper** limit  $b$ ; (ii) at the **lower** limit  $a$ ; (iii) change **inside** the interval  $(\partial f / \partial \theta)$ .

Graph on next slide: grey =  $\partial f / \partial \theta$ ; pink/blue = lower/upper limit terms  $(a'(\theta), b'(\theta))$ .



## 2.4 Leibniz rule — variable limits (graph)



## 2.4 Leibniz — practice (monopoly pricing)

**Economic setup.** A unit mass of consumers; consumer  $i$  has willingness to pay  $v_i$ , with  $v_i \sim F$  (cdf  $F$ , pdf  $f$  on  $[0, \bar{v}]$ ). A monopolist sets price  $p$ ;  $i$  buys iff  $v_i \geq p$ .

**Type-dependent cost.** Serving type  $v$  costs  $c(v)$  (e.g. quality, personalization, delivery). Profit from a buyer with valuation  $v \geq p$  is  $p - c(v)$  — margin **varies by type**.

**Total profit (no simple closed form).**

$$\pi(p) = \int_p^{\bar{v}} (p - c(v)) f(v) dv \quad (\text{cannot write as } (p - c)(1 - F(p)) \text{ when } c = c(v)).$$

The lower limit moves with  $p$  — **Leibniz is the right tool**.

**Practice.** Find  $\frac{d\pi}{dp}$  and interpret each term. If  $c$  is constant, product rule also works; here it does not.

## 2.4 Leibniz — practice solution (monopoly pricing)

**Apply variable-limits Leibniz to  $\pi(p)$  directly.** Lower limit  $a(p) = p$ , upper limit  $\bar{v}$ ; only  $p$  in the integrand is the leading  $p$ :

$$\frac{d\pi}{dp} = -(p - c(p))f(p) a'(p) + \int_p^{\bar{v}} \frac{\partial}{\partial p} [(p - c(v))f(v)] dv.$$

**Simplify.**  $a'(p) = 1$ ;  $\frac{\partial}{\partial p} [(p - c(v))f(v)] = f(v)$  (since  $c(v)$  does not depend on  $p$ ):

$$\frac{d\pi}{dp} = -(p - c(p))f(p) + \int_p^{\bar{v}} f(v) dv = -(p - c(p))f(p) + (1 - F(p)).$$

**Economic reading.**

- **Lower limit**  $-(p - c(p))f(p)$ : lose margin  $p - c(p)$  on the marginal buyer ( $v = p$ ).
- **Inside**  $1 - F(p)$ : raising  $p$  still adds  $1 \cdot dp$  profit on **each** remaining buyer.

**FOC.**  $(p - c(p))f(p) = 1 - F(p)$ ; constant  $c$  gives the usual markup rule  $p - c = \frac{1 - F(p)}{f(p)}$ .

## 2.4 Fubini's theorem

**Statement.** If  $f$  is **integrable** on region  $D$ , i.e.

$$\iint_D |f(x, y)| \, dx \, dy < \infty,$$

then

$$\iint_D f(x, y) \, dx \, dy = \int \left( \int f(x, y) \, dx \right) dy = \int \left( \int f(x, y) \, dy \right) dx,$$

(inner limits depend on the shape of  $D$ ).

**Camp shortcut.** If  $D$  is bounded and  $f$  is **continuous** on  $D$ , then  $f$  is integrable and Fubini applies.

Tonelli (for interest): if  $f \geq 0$  and measurable, the same swap holds without checking  $\iint |f|$  first.

## 2.4 Fubini — triangle example

**Example (triangle).**

$$D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq x\}.$$

Compute  $\iint_D (x + y) \, dx \, dy$ .

**y first, then x:** for fixed  $x$ ,  $y \in [0, x]$ :

$$\int_0^1 \int_0^x (x + y) \, dy \, dx = \int_0^1 \left[ xy + \frac{y^2}{2} \right]_0^x dx = \int_0^1 \frac{3x^2}{2} \, dx = \frac{1}{2}.$$

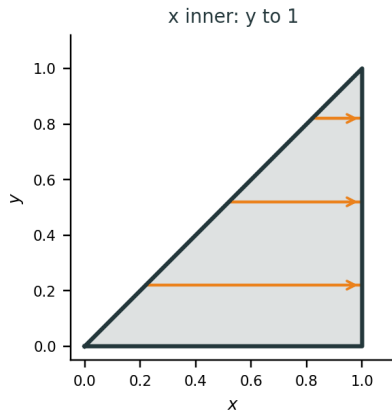
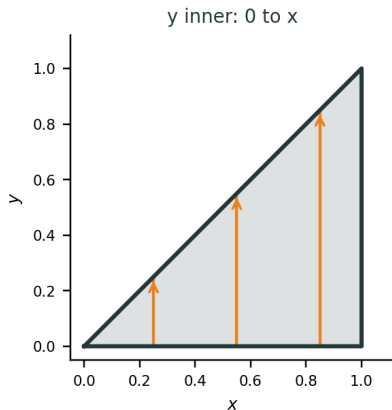
**x first, then y:** for fixed  $y$ ,  $x \in [y, 1]$ :

$$\int_0^1 \int_y^1 (x + y) \, dx \, dy = \int_0^1 \left[ \frac{x^2}{2} + xy \right]_y^1 dy = \int_0^1 \left( \frac{1}{2} + y - \frac{3y^2}{2} \right) dy = \frac{1}{2}.$$

Graph on next slide: same triangle, different slice directions.

## 2.4 Fubini — triangle (graph)

Triangle  $D$ :  $0 \leq x \leq 1$ ,  $0 \leq y \leq x$



## 2.4 Fubini — practice (between two curves)

**Region (non-rectangular).** Between the parabola  $y = x^2$  and the line  $y = x$  on  $[0, 1]$ :

$$D = \{(x, y) : 0 \leq x \leq 1, x^2 \leq y \leq x\}.$$

(On  $(0, 1)$ , the line lies **above** the parabola.)

**Practice.** Compute  $\iint_D xy \, dx \, dy$  using Fubini **twice** (different inner variable).

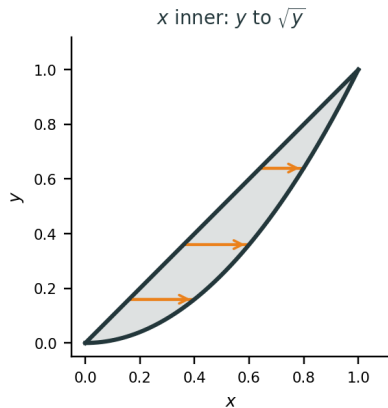
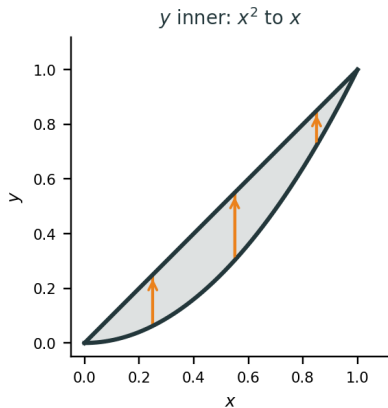
**Slice limits to find.**

- **y inner:** for fixed  $x \in [0, 1]$ ,  $y$  runs from  $x^2$  to  $x$ .
- **x inner:** for fixed  $y \in [0, 1]$ , solve  $x^2 \leq y \leq x$  to get  $x \in [y, \sqrt{y}]$ .

Graph on next slide; detailed solution on the following slides.

## 2.4 Fubini — practice (graph)

Region  $D$ :  $x^2 \leq y \leq x$  on  $0 \leq x \leq 1$





## 2.4 Fubini — practice solution ( $y$ inner first)

**Order 1:**  $\int_0^1 \int_{x^2}^x xy \, dy \, dx.$

**Inner integral (fix  $x$ ).**

$$\int_{x^2}^x xy \, dy = x \left[ \frac{y^2}{2} \right]_{y=x^2}^{y=x} = x \left( \frac{x^2}{2} - \frac{x^4}{2} \right) = \frac{x^3}{2} - \frac{x^5}{2}.$$

**Outer integral.**

$$\int_0^1 \left( \frac{x^3}{2} - \frac{x^5}{2} \right) dx = \left[ \frac{x^4}{8} - \frac{x^6}{12} \right]_0^1 = \frac{1}{8} - \frac{1}{12} = \frac{1}{24}.$$

## 2.4 Fubini — practice solution (x inner first)

**Order 2:**  $\int_0^1 \int_y^{\sqrt{y}} xy \, dx \, dy.$

**Why**  $x \in [y, \sqrt{y}]$ ? For fixed  $y$ , need  $y \leq x$  (from  $y \leq x$ ) and  $x^2 \leq y$  (from  $y \geq x^2$ ), i.e.  $x \leq \sqrt{y}$ .

**Inner integral (fix  $y$ ).**

$$\int_y^{\sqrt{y}} xy \, dx = y \left[ \frac{x^2}{2} \right]_{x=y}^{x=\sqrt{y}} = y \left( \frac{y}{2} - \frac{y^2}{2} \right) = \frac{y^2}{2} - \frac{y^3}{2}.$$

**Outer integral.**

$$\int_0^1 \left( \frac{y^2}{2} - \frac{y^3}{2} \right) dy = \left[ \frac{y^3}{6} - \frac{y^4}{8} \right]_0^1 = \frac{1}{6} - \frac{1}{8} = \frac{1}{24}.$$

**Check.** Both orders agree — Fubini is consistent when the region and integrand are well behaved.

## 2.5 Taylor expansion — formal statement

**One variable.** If  $f$  has  $n$  derivatives at  $a$ ,

$$f(a + h) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} h^k + R_n(h), \quad R_n(h) \rightarrow 0 \text{ faster than } h^n.$$

**Several variables** ( $f: \mathbb{R}^m \rightarrow \mathbb{R}$ ,  $\mathbf{a} \in \mathbb{R}^m$ ). If  $f$  is  $C^2$  near  $\mathbf{a}$ ,

$$f(\mathbf{a} + \mathbf{h}) = f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot \mathbf{h} + \frac{1}{2} \mathbf{h}^T H_f(\mathbf{a}) \mathbf{h} + o(\|\mathbf{h}\|^2),$$

$H_f(\mathbf{a})$  = Hessian at  $\mathbf{a}$  (matrix of second partials).

Proof (one variable): [appendix \(for interest only\)](#).

## 2.5 First-order approximation

**One variable** (truncate at  $k = 1$ ):

$$f(a + h) \approx f(a) + f'(a) h.$$

**Several variables:**

$$f(\mathbf{a} + \mathbf{h}) \approx f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot \mathbf{h}.$$

## 2.5 Second-order approximation

**One variable** (truncate at  $k = 2$ ):

$$f(a + h) \approx f(a) + f'(a) h + \frac{1}{2} f''(a) h^2.$$

**Several variables:**

$$f(\mathbf{a} + \mathbf{h}) \approx f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot \mathbf{h} + \frac{1}{2} \mathbf{h}^\top H_f(\mathbf{a}) \mathbf{h}.$$

**Use.** Captures **curvature** (concavity, cross-partials); improves forecasts when the first-order term is not enough.

Itô's lemma: [appendix](#) (for interest only).

## 2.5 — First-order Taylor practice (two variables) Together

**Approximate**  $f(1.1, 1.2)$  for  $f(x, y) = x^2 + y$  using Taylor at  $(1, 1)$ ,  $\mathbf{h} = (0.1, 0.2)$ .

## 2.5 — First-order Taylor practice (two variables) Together

**Approximate**  $f(1.1, 1.2)$  for  $f(x, y) = x^2 + y$  using Taylor at  $(1, 1)$ ,  $\mathbf{h} = (0.1, 0.2)$ .

**At  $(1, 1)$ :**  $f(1, 1) = 2$ ,  $f_x = 2$ ,  $f_y = 1$ .

## 2.5 — First-order Taylor practice (two variables) Together

**Approximate**  $f(1.1, 1.2)$  for  $f(x, y) = x^2 + y$  using Taylor at  $(1, 1)$ ,  $\mathbf{h} = (0.1, 0.2)$ .

**At  $(1, 1)$ :**  $f(1, 1) = 2$ ,  $f_x = 2$ ,  $f_y = 1$ .

**First-order:**  $2 + 2(0.1) + 1(0.2) = 2.4$ .



## 2.5 — First-order Taylor practice (two variables) Together

**Approximate**  $f(1.1, 1.2)$  for  $f(x, y) = x^2 + y$  using Taylor at  $(1, 1)$ ,  $\mathbf{h} = (0.1, 0.2)$ .

**At  $(1, 1)$ :**  $f(1, 1) = 2$ ,  $f_x = 2$ ,  $f_y = 1$ .

**First-order:**  $2 + 2(0.1) + 1(0.2) = 2.4$ .

**Exact**  $f(1.1, 1.2) = 2.41$ .

## 2.5 When linearization works

**Works well when:**

- the shift is **small**
- curvature is modest ( $f''$  not huge)
- the function is differentiable

**Fails or misleads when:**

- large shifts, kinks, or non-differentiability
- strong nonlinearity (e.g. corner solutions, thresholds)

## 2.5 Two variables — matrix form unpacked ignore if you are not familiar with what's Hessian

**Setup.**  $\mathbf{a} = (a_1, a_2)$ ,  $\mathbf{h} = (h_1, h_2)$ :

$$f(\mathbf{a} + \mathbf{h}) \approx f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot \mathbf{h} + \frac{1}{2} \mathbf{h}^T H_f(\mathbf{a}) \mathbf{h}.$$

**First-order.**  $\nabla f = (f_{x_1}, f_{x_2})$  at  $\mathbf{a}$ , so  $\nabla f \cdot \mathbf{h} = f_{x_1} h_1 + f_{x_2} h_2$ .

**Second-order.**  $H_f = \begin{pmatrix} f_{x_1 x_1} & f_{x_1 x_2} \\ f_{x_2 x_1} & f_{x_2 x_2} \end{pmatrix}$ , and multiply:

$$\mathbf{h}^T H_f \mathbf{h} = (h_1 \ h_2) \begin{pmatrix} f_{x_1 x_1} h_1 + f_{x_1 x_2} h_2 \\ f_{x_2 x_1} h_1 + f_{x_2 x_2} h_2 \end{pmatrix} = f_{x_1 x_1} h_1^2 + (f_{x_1 x_2} + f_{x_2 x_1}) h_1 h_2 + f_{x_2 x_2} h_2^2.$$

If  $f \in C^2$ , then  $f_{x_1 x_2} = f_{x_2 x_1}$ , hence

$$f(\mathbf{a} + \mathbf{h}) \approx f(\mathbf{a}) + f_{x_1} h_1 + f_{x_2} h_2 + \frac{1}{2} (f_{x_1 x_1} h_1^2 + 2f_{x_1 x_2} h_1 h_2 + f_{x_2 x_2} h_2^2).$$

## 2.5 — Second-order Taylor practice (two variables)

Take home

ignore if you are

not familiar with what's Hessian

**Approximate**  $f(0.1, 0.2)$  for  $f(x, y) = \sin x + \cos y$  using a second-order Taylor expansion at  $(0, 0)$ .

Use  $\mathbf{h} = (0.1, 0.2)$  and the explicit second-order formula on the matrix-form slide.

Check your answer with a calculator (no need to simplify trig decimals).

## 2.6 Pointwise vs. uniform convergence

**Pointwise.**  $f_n \rightarrow f$  on  $D$  means: for each fixed  $x \in D$ ,

$$f_n(x) \rightarrow f(x) \quad \text{as } n \rightarrow \infty.$$

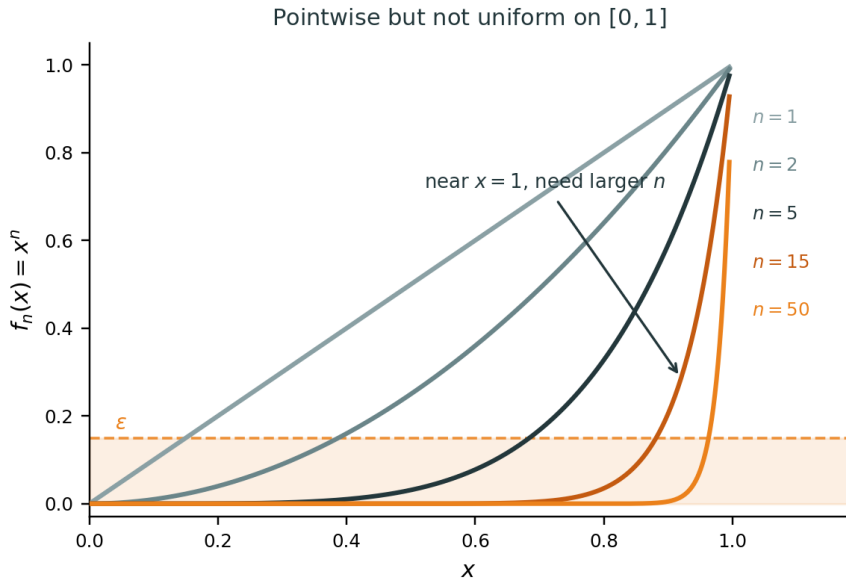
Equivalently: for each  $x$  and every  $\varepsilon > 0$ ,  $\exists N$  (may depend on **both**  $x$  **and**  $\varepsilon$ ) such that  $|f_n(x) - f(x)| < \varepsilon$  for all  $n \geq N$ .

**Uniform.** For every  $\varepsilon > 0$ ,  $\exists N$  (depends on  $\varepsilon$  **only**) such that for all  $n \geq N$  and **all**  $x \in D$ ,

$$|f_n(x) - f(x)| < \varepsilon.$$

One  $N$  works **uniformly** over the whole domain.

## 2.6 Example — pointwise but not uniform



## 2.6 Why uniform convergence matters

**Uniform convergence is the safe condition** for swapping limits with other operations:

- **Continuity:** if each  $f_n$  is continuous and  $f_n \rightarrow f$  **uniformly**, then  $f$  is continuous.
- **Integration:** under uniform convergence,  $\lim_{n \rightarrow \infty} \int f_n = \int \lim_{n \rightarrow \infty} f_n$ .
- **Differentiation** (with extra hypotheses): uniform convergence of derivatives can justify  $\frac{d}{dx} \lim f_n = \lim \frac{d}{dx} f_n$ .
- **Example** ( $x^n$  on  $[0, 1]$ ): each  $f_n$  is continuous, but the pointwise limit (0 on  $[0, 1)$ , 1 at  $x = 1$ ) jumps at  $x = 1$  — **continuity fails** without uniform convergence.

# Appendix

**Not required for Math Camp.** Proofs and extra examples linked from the main slides.



## A.1 IFT — several variables

**System:**  $\mathbf{F}(\mathbf{y}, \boldsymbol{\theta}) = \mathbf{0}$ ,  $\mathbf{F}: \mathbb{R}^{n+m} \rightarrow \mathbb{R}^n$ .

If the  $n \times n$  Jacobian  $\frac{\partial \mathbf{F}}{\partial \mathbf{y}}$  is invertible at the point, then locally  $\mathbf{y} = \mathbf{y}^*(\boldsymbol{\theta})$  and

$$\frac{\partial \mathbf{y}^*}{\partial \boldsymbol{\theta}} = - \left( \frac{\partial \mathbf{F}}{\partial \mathbf{y}} \right)^{-1} \frac{\partial \mathbf{F}}{\partial \boldsymbol{\theta}}.$$

Same pattern as the one-variable formulas with  $F_y$  and  $F_\theta$ . Appears in graduate micro; camp focus is the scalar case in section 2.1.

## A.2 Proof. EVT on $[a, b]$

for interest only

**Claim.** If  $f$  is continuous on  $[a, b]$ , then  $f$  attains a maximum and minimum on  $[a, b]$ .

**Bolzano–Weierstrass Theorem.** Every bounded sequence in  $\mathbb{R}$  has a convergent subsequence.

**Step 1.**  $f$  is bounded on  $[a, b]$ . Assume, for contradiction, that  $f$  is unbounded. Then  $\exists (x_n)$  in  $[a, b]$  with  $|f(x_n)| > n$  for all  $n$ . The sequence  $(x_n)$  is bounded (it lies in  $[a, b]$ ). By Bolzano–Weierstrass, some subsequence  $(x_{n_k})$  converges to  $c \in [a, b]$ . Continuity gives  $f(x_{n_k}) \rightarrow f(c)$ . But  $|f(x_{n_k})| > n_k \rightarrow \infty$ , a contradiction. Hence  $f$  is bounded; let  $M = \sup\{f(x) : x \in [a, b]\}$ .

## A.2 Proof. EVT on $[a, b]$ (continued)

**Step 2 — claim.**  $f$  attains its maximum  $M = \sup\{f(x) : x \in [a, b]\}$  on  $[a, b]$ ; i.e.  $\exists d \in [a, b]$  with  $f(d) = M$ .

**Why  $\exists (y_n)$  in  $[a, b]$  with  $f(y_n) \rightarrow M$ .** By definition of supremum:

1.  $f(x) \leq M$  for all  $x \in [a, b]$ ;
2. for every  $\varepsilon > 0$ ,  $\exists x \in [a, b]$  with  $f(x) > M - \varepsilon$ .

Take  $\varepsilon_n = \frac{1}{n}$ : pick  $y_n \in [a, b]$  with  $f(y_n) > M - \frac{1}{n}$ . Since  $f(y_n) \leq M$ ,

$$M - \frac{1}{n} < f(y_n) \leq M \quad \Rightarrow \quad f(y_n) \rightarrow M.$$

## A.2 Proof. EVT on $[a, b]$ (continued)

Step 2 — finish.

1.  $(y_n)$  is **bounded**: each  $y_n \in [a, b] \subset \mathbb{R}$ .
2. By Bolzano–Weierstrass, some subsequence  $(y_{n_k})$  converges to a limit  $d \in [a, b]$ .
3. Since  $f(y_n) \rightarrow M$ , every subsequence has the **same limit**:

$$f(y_{n_k}) \rightarrow M.$$

4. By **continuity** of  $f$  at  $d$ ,

$$f(d) = \lim_{k \rightarrow \infty} f(y_{n_k}) = M.$$

So  $M$  is attained at  $d \in [a, b]$ . Minimum: repeat for  $-f$ .



## A.3 Expectation from the CDF

**Claim.** For non-negative continuous  $X$  with PDF  $f$  and CDF  $F$ ,

$$\mathbb{E}[X] = \int_0^{\infty} xf(x) dx = \int_0^{\infty} (1 - F(x)) dx.$$

**Proof (integration by parts).** Set  $u = 1 - F(x)$  and  $dv = dx$ . Then  $du = -f(x) dx$  and  $v = x$ :

$$\int_0^{\infty} (1 - F(x)) dx = [x(1 - F(x))]_0^{\infty} - \int_0^{\infty} x(-f(x)) dx = [x(1 - F(x))]_0^{\infty} + \int_0^{\infty} xf(x) dx.$$

At  $x = 0$ ,  $x(1 - F(x)) = 0$ . As  $x \rightarrow \infty$ ,  $x(1 - F(x)) \rightarrow 0$  when  $\mathbb{E}[X] < \infty$ . Hence

$$\int_0^{\infty} (1 - F(x)) dx = \mathbb{E}[X].$$

□

## A.4 Leibniz rule — failure example

**Example (fixed limits).** On  $[0, 1]$ , let

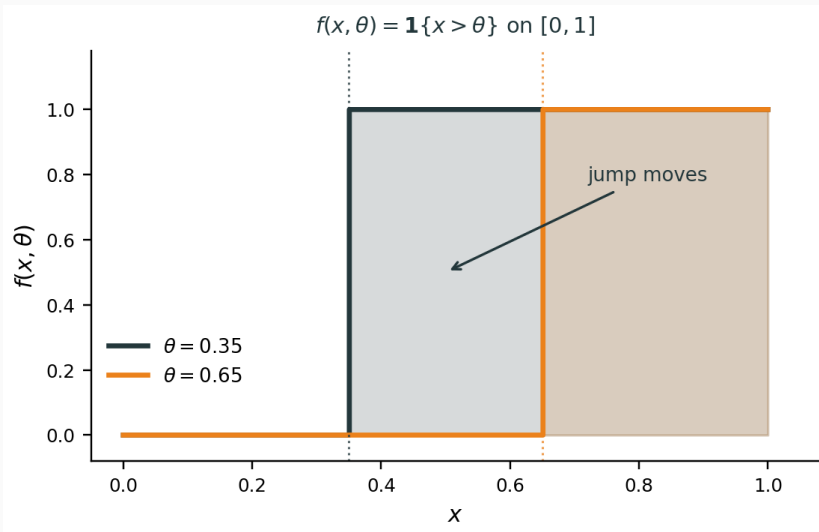
$$f(x, \theta) = \mathbf{1}\{x > \theta\} \quad (\text{step jumps at } x = \theta).$$

Then

$$F(\theta) = \int_0^1 f(x, \theta) dx = 1 - \theta \quad \Rightarrow \quad F'(\theta) = -1.$$

**Why Leibniz fails.** The discontinuity **moves with**  $\theta$ , so  $\partial f / \partial \theta$  is not continuous on  $[0, 1] \times I$ . Naively  $\partial f / \partial \theta = 0$  a.e., so  $\int_0^1 0 dx = 0 \neq F'(\theta)$ .

## A.4 Leibniz failure — moving step (graph)



## A.5 Taylor expansion (one variable)

**Claim (second order).** If  $f \in C^2$  on an interval containing  $a$  and  $a + h$ , then

$$f(a + h) = f(a) + f'(a)h + \frac{1}{2}f''(c)h^2$$

for some  $c$  strictly between  $a$  and  $a + h$ .

**Proof.** Write

$$f(a + h) = f(a) + hf'(a) + R(h).$$

**Step 1.** By FTC and the chain rule,  $f(a + h) - f(a) = \int_0^h f'(a + t) dt$ , so

$$R(h) = \int_0^h [f'(a + t) - f'(a)] dt.$$

**Step 2.** For each  $t \in (0, h]$ , MVT on  $f'$  gives  $f'(a + t) - f'(a) = tf''(c_t)$  with  $c_t \in (a, a + t)$ . Hence

$$R(h) = \int_0^h tf''(c_t) dt = \frac{h^2}{2} f''(c)$$

for some  $c$  between  $a$  and  $a + h$  (continuous  $f''$ ). □

First order follows from the  $n = 1$  case (or MVT on  $f$  directly). Multivariable formulas follow from applying the chain rule along  $\mathbf{a} + t\mathbf{h}$ .



## A.6 What is Brownian motion?

**Brownian motion**  $W_t$  (also called a **Wiener process**): a continuous-time **random** path with  $W_0 = 0$ .

**Intuition.** Think of a particle jolted by unpredictable shocks (e.g. noise in a stock price). You cannot forecast the next jiggle, but you can describe its **typical size**.

**Key facts (informal).**

- Over  $[t, t + \Delta t]$ : increment  $\Delta W = W_{t+\Delta t} - W_t$  has mean 0 and variance  $\Delta t$
- Increments on **non-overlapping** intervals are independent
- Paths are **continuous** but very jagged (not differentiable in the usual sense)

In  $dX_t = \mu dt + \sigma dW_t$ , the  $dW_t$  term is this random noise scaled by  $\sigma$ .

## A.6 Taylor and Itô's lemma

**Setup.**  $X_t$  follows an Itô process (e.g.  $dX_t = \mu dt + \sigma dW_t$ ). Over a short interval  $[t, t + \Delta t]$ ,

$$\Delta X = X_{t+\Delta t} - X_t \quad (\text{change in the state variable}).$$

Expand  $f(X_t, t)$  in the **two** small steps  $\Delta X$  and  $\Delta t$ .

**Taylor in  $(x, t)$  through order 2:**

$$\Delta f \approx f_x \Delta X + f_t \Delta t + \frac{1}{2} f_{xx} (\Delta X)^2 + f_{xt} \Delta X \Delta t + \frac{1}{2} f_{tt} (\Delta t)^2.$$

## A.6 Itô multiplication rules

**Ordinary calculus.** If  $\Delta t$  is small, then  $(\Delta t)^2$  is **much smaller** and can be dropped.

**Stochastic calculus.**  $\Delta W = W_{t+\Delta t} - W_t$  is one **Brownian increment** (see previous slide): mean 0, variance  $\Delta t$ . So a typical size is  $|\Delta W| \sim \sqrt{\Delta t}$ , not  $\Delta t$ .

**Why**  $(\Delta W)^2 \approx \Delta t$ . Squaring a  $\sqrt{\Delta t}$ -sized step gives order  $\Delta t$ :

$$(\Delta W)^2 \sim (\sqrt{\Delta t})^2 = \Delta t.$$

So  $(\Delta W)^2$  is **not** negligible next to  $\Delta t$  (unlike  $(\Delta t)^2$ ).

**Itô rules (informally):**  $(\Delta t)^2 \rightarrow 0$ ,  $(\Delta W)^2 \approx \Delta t$ ,  $\Delta t \Delta W \rightarrow 0$ . For  $dX = \mu dt + \sigma dW$ , this makes  $\frac{1}{2} f_{xx} (\Delta X)^2$  contribute at the same scale as  $\Delta t$ .

**Itô's lemma:**

$$df = f_t dt + f_x dX + \frac{1}{2} f_{xx} (dX)^2 \quad (\text{plus higher-order terms}).$$